**Remark**

(1) An angle between two non-zero vectors $\vec{a}$ and $\vec{b}$ is found by the following formula

$$\theta = \cos^{-1}\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}\right).$$

(2) $\vec{a}$ and $\vec{b}$ are said to be parallel if the angle between them is $0$ or $\pi$.

(3) $\vec{a}$ and $\vec{b}$ are said to be perpendicular if the angle between them is $\frac{\pi}{2}$ or $\frac{3\pi}{2}$.

**Property**

(1) Let $\vec{a}$ and $\vec{b}$ be any two nonzero vectors. Then

- $\vec{a}\cdot\vec{b}=0$ if and only if $\vec{a}$ and $\vec{b}$ are perpendicular to each other.
- $\vec{a}\times\vec{b}=\vec{0}$ if and only if $\vec{a}$ and $\vec{b}$ are parallel to each other.

(2) If $\vec{a},\vec{b}$, and $\vec{c}$ are any three vectors and $\alpha$ is a scalar, then

$$\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a},\ (\vec{a}+\vec{b})\cdot\vec{c}=\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c},\ (\alpha\vec{a})\cdot\vec{b}=\alpha(\vec{a}\cdot\vec{b})=\vec{a}\cdot(\alpha\vec{b});$$

$$\vec{a}\times\vec{b} = -(\vec{b}\times\vec{a}),\ (\vec{a}+\vec{b})\times\vec{c}=\vec{a}\times\vec{c}+\vec{b}\times\vec{c},\ (\alpha\vec{a})\times\vec{b}=\alpha(\vec{a}\times\vec{b})=\vec{a}\times(\alpha\vec{b}).$$

## 6.3.2 Application of dot and cross products in plane Trigonometry

We apply the concepts of dot and cross products of two vectors to derive a few formulae in plane trigonometry.

**Example 6.1 (Cosine formulae)**

With usual notations, in any triangle *ABC*, prove the following by vector method.

(i) $a^2=b^2+c^2-2bc\cos A$ (ii) $b^2=c^2+a^2-2ca\cos B$

(iii) $c^2=a^2+b^2-2ab\cos C$

**Solution**

With usual notations in triangle *ABC*, we have $\overrightarrow{BC}=\vec{a},\overrightarrow{CA}=\vec{b}$ and $\overrightarrow{AB}=\vec{c}$. Then $|\overrightarrow{BC}|=a, |\overrightarrow{CA}|=b$, $|\overrightarrow{AB}|=c$ and $\overrightarrow{BC}+\overrightarrow{CA}+\overrightarrow{AB} = \vec{0}$.

So, $\overrightarrow{BC} = -\overrightarrow{CA}-\overrightarrow{AB}$.

Then applying dot product, we get

$$\overrightarrow{BC}\cdot\overrightarrow{BC} = (-\overrightarrow{CA}-\overrightarrow{AB})\cdot(-\overrightarrow{CA}-\overrightarrow{AB})$$

$$\Rightarrow\ |\overrightarrow{BC}|^2 = |\overrightarrow{CA}|^2+|\overrightarrow{AB}|^2+2\overrightarrow{CA}\cdot\overrightarrow{AB}$$

$$\Rightarrow\ a^2 = b^2+c^2+2bc\cos(\pi-A)$$

$$\Rightarrow\ a^2 = b^2+c^2-2bc\cos A.$$

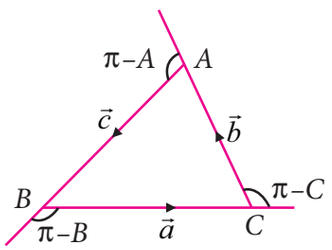


Fig. 6.6

The results in (ii) and (iii) are proved in a similar way. ■

**Example 6.2**

With usual notations, in any triangle *ABC*, prove the following by vector method.

(i) $a=b\cos C+c\cos B$ (ii) $b=c\cos A+a\cos C$

(iii) $c=a\cos B+b\cos A$



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